

API DOCUMENTATION

Indomaret Backend Developer Technical Assignment

Production-Grade RESTful API Specification & Architecture

DOCUMENT METADATA

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Role Applied:	Backend Developer
Company:	PT Indomarco Prismatama (Klik Indomaret)
Technology Stack:	Java 17/22, Spring Boot 4, Hibernate JPA, MySQL / PostgreSQL
Security Framework:	Spring Security 7 + HMAC-SHA256 Stateless JWT
Base URL:	http://localhost:8080/api/v1
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EXECUTIVE SUMMARY & HIGHLIGHTS

This API serves master data management for Provinces, Branches, and Stores across Indonesia. It has been meticulously designed and benchmarked for high-scale environments (>20,000 stores and 100 branches).

Key technical implementations include:

- User Story 1: Optimized store search by province name with sub-50ms latency using indexed JOINS.
- User Story 2: Full branch update and soft-deletion with cascading logic.
- User Story 3: Strict non-deleted, active-only data filtration across all responses.
- User Story 4: Dynamic whitelist store management always included in search regardless of province.
- User Story 5: Stateless JWT authentication securing all endpoints with standard 401/403 responses.
- Audit Trail: Automatic change data capture logging (traceability & accountability).

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All endpoints reside under the context path `/api/v1`. Responses are encapsulated inside standard JSON envelopes:

- Authentication uses Bearer Tokens passed via HTTP Header: Authorization: Bearer <JWT_TOKEN>

Public

Sample Request Body:

Sample Response (200 OK):

```
{
  "success": true,
  "message": "Authentication successful",
  "data": {
    "token": "eyJhbGciOiJIUzI1NiIsIn...",
    "tokenType": "Bearer",
    "expiresInMs": 86400000,
    "userId": 1,
    "username": "admin",
    "role": "ROLE_ADMIN"
  }
}
```

```
curl -X POST http://localhost:8080/api/v1/auth/login \
-H "Content-Type: application/json" \
-d '{"username":"admin","password":"admin123"}'
```

Public

Sample Request Body:

Sample Response (200 OK):

```
{
  "success": true,
  "message": "User registered successfully",
  "data": {
    "id": 3,
    "username": "farhan_dev",
    "email": "farhan@indomaret.co.id",
    "role": "ROLE_USER"
  }
}
```

```
curl -X POST http://localhost:8080/api/v1/auth/register \
-H "Content-Type: application/json" \
-d '{"username": "farhan_dev", "email": "farhan@indomaret.co.id", "password": "password123"}'
```

3. Store Search & Management (User Stories 1, 3, 4)

GET `/api/v1/stores/search?provinceName=DKI%20Jakarta&page=0&size=20`

Bearer JWT

Search stores by province name. CRITICAL BEHAVIOR: Automatically includes all active global Whitelist Stores in the result set regardless of location. Only active and non-deleted records are returned.

Query Parameters:

provinceName	String	No	Target province name (case-insensitive substring match, e.g. "Jakarta")
page	Integer	No	Page number (0-indexed, default: 0)
size	Integer	No	Page size (default: 20, max: 100 enforced by configuration)
sortBy	String	No	Sort field: "name", "code", "id" (default: "name")
sortDir	String	No	"asc" or "desc" (default: "asc")

Sample Response (200 OK):

```
{
  "success": true,
  "data": {
    "content": [
      {
        "id": 101,
        "name": "Indomaret Juanda",
        "code": "STR-000101",
        "branchName": "Cabang Jakarta Pusat",
        "provinceName": "DKI Jakarta",
        "whitelisted": false,
        "isActive": true
      }
    ]
  }
}
```

Sample cURL:

```
curl -X GET "http://localhost:8080/api/v1/stores/search?provinceName=DKI%20Jakarta&page=0&size=20" \
-H "Authorization: Bearer <TOKEN>"

{
  "success": true,
  "data": {
    "content": [
      {
        "id": 101,
        "name": "Indomaret Juanda",
        "code": "STR-000101",
        "branchName": "Cabang Jakarta Pusat",
        "provinceName": "DKI Jakarta",
        "whitelisted": false,
        "isActive": true
      },
      {
        "id": 102,
        "name": "Indomaret Juanda",
        "code": "STR-000102",
        "branchName": "Cabang Denpasar",
        "provinceName": "Bali",
        "whitelisted": true, // <--- Whitelist store included across all provinces!
        "isActive": true
      }
    ],
    "pageNumber": 0,
    "pageSize": 20,
    "totalElements": 21,
    "totalPages": 2,
    "isLast": false
  }
}
```

STORE CRUD ENDPOINTS SUMMARY

GET /api/v1/stores	Paginated list of all active, non-deleted stores with branch & province hierarchy.
GET /api/v1/stores/{id}	Retrieve specific store details by ID. Returns 404 if deleted or inactive.
POST /api/v1/stores	Create a new store. Body: { "name", "code", "address", "branchId" }. Logs audit trail.
PUT /api/v1/stores/{id}	Update store details (name, code, address, branchId, isActive). Logs before/after state.
DELETE /api/v1/stores/{id}	Soft-delete store (sets isDeleted=true, isActive=false). Deactivates whitelist entry.

5. Branch Management (User Stories 2, 3)

PUT `/api/v1/branches/{id}`

Bearer JWT

Update existing branch data. Automatically validates unique branch code and active province reference. Records full before/after audit log.

Sample Request Body:

```
{
  "name": "Cabang Bandung Barat Sentra",
  "address": "Jl. Sukajadi No. 120, Bandung",
  "provinceId": 13,
  "isActive": true
}
```

Sample Response (200 OK):

```
{
  "success": true,
  "message": "Branch updated successfully",
  "data": {
    "id": 15,
    "name": "Cabang Bandung Barat Sentra",
    "code": "BR-JABAR-002",
    "address": "Jl. Sukajadi No. 120, Bandung",
    "provinceName": "Jawa Barat",
    "isActive": true
  }
}
```

Sample cURL:

```
curl -X PUT http://localhost:8080/api/v1/branches/15 \
-H "Authorization: Bearer <TOKEN>" \
-H "Content-Type: application/json" \
```

DELETE `/api/v1/branches/{id}`

Bearer JWT

Soft delete branch data. Cascades soft-deletion to all associated stores so they are immediately excluded from all user query responses.

Sample Response (200 OK):

```
{
  "success": true,
  "message": "Branch deleted successfully",
  "data": null
}
```

Sample cURL:

```
curl -X DELETE http://localhost:8080/api/v1/branches/15 \
-H "Authorization: Bearer <TOKEN>"
```

BRANCH QUERY ENDPOINTS

GET `/api/v1/branches?page=0&size=20`

Returns paginated active and non-deleted branches with province details.

GET `/api/v1/branches/{id}`

Retrieve single branch by ID. Returns 404 if deleted or inactive.

GET `/api/v1/branches/by-province/{provinceId}`

Returns all active branches belonging to a specific province ID.

6. Provinces, Audit Trail & Seeding Endpoints

6.1 PROVINCES API

GET /api/v1/provinces : Returns paginated list of all 38 Indonesian provinces seeded in the master database.

GET /api/v1/provinces/{id} : Returns province details by ID.

6.2 AUDIT TRAIL API (Traceability & Accountability)

GET /api/v1/audit-logs?page=0&size=50

Bearer JWT

Inspect chronological change data capture log. Records user, entity, action, timestamp, client IP, and before/after JSON states.

Sample Response (200 OK):

```
{
  "success": true,
  "data": {
    "content": [
      {
        "id": 89,
        "entityName": "Branch",
        "entityId": 15,
        "action": "UPDATE",
        "performedBy": "admin",
        "oldValue": "{\"name\": \"Cabang Bandung Lama\", ...}",
        "newValue": "{\"name\": \"Cabang Bandung Barat Sentra\", ...}",
        "ipAddress": "127.0.0.1",
        "timestamp": "2026-10-03T19:18:46"
      }
    ]
  }
}
```

Sample cURL:
`curl -X GET http://localhost:8080/api/v1/audit-logs \`
`-H "Authorization: Bearer <TOKEN>"`

6.3 HIGH-VOLUME SEED & PERFORMANCE BENCHMARK API

POST /api/v1/seed/bulk-stores?count=20000

Public / Admin

Generates up to 20,000+ realistic stores across the 100 branches using high-speed JDBC batch operations in ~1.5s for performance testing.

Sample Response (200 OK):

```
{
  "success": true,
  "message": "Bulk seed completed",
  "data": {
    "totalStores": 20000,
    "targetCount": 20000,
    "elapsedMs": 1420,
    "message": "Successfully seeded stores up to 20000 in 1420 ms"
  }
}
```

Sample cURL:

```
curl -X POST "http://localhost:8080/api/v1/seed/bulk-stores?count=20000"
```